Course outline

1. Linear ordinary differential equations (ODEs) and initial value problems (IVPs)
2. Power series solutions to ODEs
3. Laplace transforms
4. Fourier series
5. Boundary value problems (BVPs) and linear partial differential equations (PDEs)
6. Fourier transforms

Topic 2: Power series solutions to ODEs

How Topic 2 builds on Topic 1:

- The solutions to the ODEs in Topic 1 involve familiar "elementary" functions that we have seen countless times in pre-requisite subjects (e.g., polynomials, exponentials, trigonometric functions, etc.)
- In Topic 2, we will encounter a richer class of ODEs whose solutions cannot be expressed in terms of these elementary functions. Instead they give rise to power series solutions which are classed under "Special Functions".

Topic 2: Power series solutions to ODEs

Leaning outcomes:

1. Discover how power series and associated ODEs arise in real-world applications
2. Learn power series method for solving ODEs and IVPs
3. Where does a given series converge? To which function does a series converge?

We will:

1. Revise the concept of a power series and calculate power series for some simple functions
2. Motivate the use of power series for solving ODEs
3. Revise the ratio test for determining the radius of convergence of a power series

Topic 2: Power Series Solutions to ODEs

- You are familiar with the concept of a series, e.g.,

$$\sum_{n=0}^{\infty} c_n \ ,$$

  how to test if they converge and what they converge to.

- These series may be functions of t, e.g., a power series

$$\sum_{n=0}^{\infty} c_n \, (t-a)^n \ .$$

  In this case, whether the series converges or not will generally depend on t.

Definition (Taylor series)

- If $f(t)$ is N-times differentiable, the Nth-order Taylor polynomial around $t = a$ is

$$P_N(t) = \sum_{n=0}^{N} \frac{f^{(n)}(a)}{n!} (t-a)^n \ .$$

  For $N > 1$, this generalises the linear approximation of a function

- If $f(t)$ is infinitely differentiable ("smooth"), then the Taylor series of f around a is

$$P(t) = \lim_{N\to\infty} P_N(t) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (t-a)^n \qquad \text{(if the limit exists)}$$

  The function $P(t)$ may (or may not) agree with the function $f(t)$

- If $a = 0$ then it is often known as a Maclaurin series

Exercise 2.1: Find the Taylor series of $f(t) = \mathrm{e}^t$ around $t = 0$

The derivatives are

$$f^{(n)}(t) = \mathrm{e}^t \quad \text{for} \quad n = 0, 1, 2, \ldots$$

$$\Rightarrow \quad f^{(n)}(0) = 1 \quad \text{for} \quad n = 0, 1, 2, \ldots$$

Thus, the Taylor series for e^t around $t = 0$ is

$$P(t) = \sum_{n=0}^{\infty} \frac{t^n}{n!} = 1 + t + \frac{1}{2}t^2 + \cdots$$

Exercise 2.2: Find the Taylor series of $f(t) = \dfrac{1}{1-t}$ about $t = 0$

$$f(t) = \frac{1}{1-t} \quad \Rightarrow \quad f(0) = 1$$

$$f'(t) = \frac{1}{(1-t)^2} \quad \Rightarrow \quad f'(0) = 1$$

$$f''(t) = \frac{2}{(1-t)^3} \quad \Rightarrow \quad f''(0) = 2$$

$$f'''(t) = \frac{6}{(1-t)^4} \quad \Rightarrow \quad f'''(0) = 6$$

$$f^{(n)}(t) = \frac{n!}{(1-t)^{n+1}} \quad \Rightarrow \quad f^{(n)}(0) = n!$$

Use this to find the Taylor series

$$P(t) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}(t-0)^n = \sum_{n=0}^{\infty} \frac{n!}{n!}t^n = \sum_{n=0}^{\infty} t^n$$

As this is a geometric series, it converges for $|t| < 1$

In this case, the function $f(t)$ can be represented by its Maclaurin series and we have

$$\frac{1}{1-t} = \sum_{n=0}^{\infty} t^n \qquad \text{for } |t| < 1$$

Note that this identity fails for $|t| > 1$ as the right hand side does not converge.

Brain teaser: What replaces the right hand side for $|t| > 1$?

Exercise 2.3: Solve y' = y

The answer is $y(t) = C\,\mathrm{e}^t$ ... buuttt$_{\mathrm{ttt}}$

... what if the exponential function hadn't been discovered?

Try a Maclaurin series of form

$$y(t) = \sum_{n=0}^{\infty} c_n\, t^n$$

$$\Rightarrow \quad y'(t) = \sum_{n=1}^{\infty} n\, c_n\, t^{n-1} = \sum_{m=0}^{\infty} (m+1)\, c_{m+1}\, t^m \quad \text{making an assumption!}$$

$$\Rightarrow \quad \underbrace{\sum_{m=0}^{\infty} (m+1)\, c_{m+1}\, t^m}_{y'} = \underbrace{\sum_{m=0}^{\infty} c_m\, t^m}_{y} \quad \Rightarrow \quad \sum_{m=0}^{\infty} \Big\{ (m+1)\, c_{m+1} - c_m \Big\}\, t^m = 0$$

Therefore, require

$$(m+1)\,c_{m+1} - c_m = 0 \quad \text{for} \quad m = 0, 1, 2, \ldots$$

$$c_{m+1} = \frac{c_m}{m+1} \quad \text{for} \quad m = 0, 1, 2, \ldots$$

a recurrence relation.

Let $c_0 = C$, then

$$c_1 = \frac{C}{1}\,, \quad c_2 = \frac{c_1}{2} = \frac{C}{2 \cdot 1}\,, \quad c_3 = \frac{c_2}{3} = \frac{C}{3 \cdot 2 \cdot 1}\,, \ldots, \; c_n = \frac{C}{n!}$$

Our solution is

$$y(t) = \sum_{n=0}^{\infty} c_n\, t^n = C \sum_{n=0}^{\infty} \frac{t^n}{n!} = C\, \mathrm{e}^t \qquad \text{for all } t$$

If we did not know how to evaluate the series, the last step could have been inferred from the uniqueness of the IVP assuming $y(0) = 1$

Reminder: Convergence of a series via ratio test

Theorem (Ratio test for series)

For a series $\sum_{n=0}^{\infty} a_n$ with $a_n \neq 0$ for all large enough n compute

$$\ell = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| .$$

- The series converges for $\ell < 1$ and diverges for $\ell > 1$.
- For $\ell = 1$ the test is inconclusive.
- In that case convergence needs to be checked by different means (e.g. the root test)

Exercise 2.4: For what values of t does the series $\displaystyle\sum_{n=1}^{\infty} \frac{t^n}{2n}$ converge?

We have

$$c_0 = 0 \quad \text{and} \quad c_n = \frac{1}{2n} \text{ for } n > 0$$

Apply ratio test

$$\ell = \lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n} t \right| = \lim_{n\to\infty} \left| \frac{2n}{2(n+1)} t \right| = \lim_{n\to\infty} \left(\frac{1}{1+\frac{1}{n}} \right) |t| = |t| < 1$$

We will later refer to R in $|t| < R = 1$ as the radius of convergence

Convergence of power series – part I

There are two general questions to answer about power series $P(t) = \sum_{n=0}^{\infty} c_n (t-a)^n$.

- For which t does the power series converge?
- When this power series converges, what does it converge to?

The first question is usually answered using an adaption of the ratio test to power series.

In this case $a_n = c_n (t-a)^n$ and the ratio test for general series demands us to compute

$$\ell = \lim_{n\to\infty} \left| \frac{c_{n+1}(t-a)^{n+1}}{c_n (t-a)^n} \right| = |t-a|\, L \qquad \text{with} \qquad L = \lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n} \right| .$$

We thus have convergence for $|t-a| < L^{-1}$ and divergence for $|t-a| > L^{-1}$.

Convergence of power series – part II

The previous result motivates to define the radius of convergence $R = L^{-1}$, in terms of which we find

Theorem (Ratio test for power series)

For a power series $P(t) = \sum_{n=0}^{\infty} c_n (t-a)^n$ with $c_n \neq 0$ for all large enough n compute

$$R = \lim_{n\to\infty} \left| \frac{c_n}{c_{n+1}} \right| = L^{-1} \qquad \text{(note the opposite order in the quotient!)}$$

- The power series converges for $|t-a| < R$ and diverges for $|t-a| > R$
- For $|t-a| = R$ one has to check carefully using another test

Note that $R = 0$ and $R = \infty$ are permitted for the purposes of this test. A power series always converges for $t = a$.

Exercise 2.5: Use the ratio test on the power series

$$P(t) = \sum_{n=1}^{\infty} c_n\, t^n \quad \text{where} \quad c_n = \begin{cases} \frac{1}{2^n} & (n \text{ even}) \\ 0 & (n \text{ odd}) \end{cases}$$

i.e., every other term is zero.

The idea is to apply ratio test to sequential non-zero terms $n = 2k$ and $2(k+1)$.

In practice, we set $n = 2k$ for $k = 1, 2, \ldots$ and rewrite the series as

$$P(t) = \sum_{n=1}^{\infty} c_n\, t^n = \sum_{k=1}^{\infty} \frac{t^{2k}}{2^{2k}} = \sum_{k=1}^{\infty} \left(\frac{t^2}{4}\right)^k = \frac{1}{1 - \frac{t^2}{4}} - 1 \qquad \text{for} \quad |t| < 2$$

Applying the general ratio test to this series leads to the condition

$$\lim_{k\to\infty} \left| \frac{t^{2(k+1)}}{2^{2(k+1)}} \frac{2^{2k}}{t^{2k}} \right| < 1 \qquad \Rightarrow \qquad \left| \frac{t^2}{4} \right| < 1 \quad \Rightarrow \quad |t| < 2$$

Exercise 2.6: Taylor series don't always converge to the original function

Consider the function $f(t) = \begin{cases} e^{-\frac{1}{t^2}} & t \neq 0 \\ 0 & t = 0 \end{cases}$

- $f(t)$ is continuous and infinitely differentiable ("smooth") for all t
- $f^{(n)}(0) = 0$ for all n
- Thus, the Taylor series of $f(t)$ around $a = 0$ is

$$P(t) = \sum_{n=0}^{\infty} \frac{0}{n!} t^n = 0 \quad \text{for all } t$$

